

Pranav Munigala

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EDUCATION

Rutgers University – Engineering School

New Brunswick, NJ

Bachelor of Science (B.S.) in Biomedical Engineering

Expected Graduation Date: May 2028

- **GPA:** 4.0
- **Honors:** Honors College | Dean's List
- **Relevant Coursework:** Honors Multivariable Calculus, Analytical Physics 1, Honors Engineering Mechanics Statics, Intro to Linear Algebra, Honors IDEA 1 & 2, Differential Equations, Intro to Mathematical Reasoning

WORK & LEADERSHIP EXPERIENCE

Inferyx: AI Solutions Company

Basking Ridge, NJ

Data Analytics Intern

June 2024 – January 2025

- Led an independent bioinformatics research project using Python, Biopython, ChEMBL, and PDB data to extract, filter, and analyze 3D structure metrics, molecular properties, and bioactivity metrics and developed a Streamlit tool that visualizes protein structures and answers user-driven structure/function questions via an AI agent.
- Presented and defended data-driven bioinformatics application proposals to company, demonstrating how compound and protein metrics can bridge Inferyx's new solutions with the emerging biomedical industry needs.

Main Street Medical

Parsippany, NJ

Intern/Shadow

August 2024 – February 2025

- Supported clinical operations by organizing and converting patient intake forms into electronic records, updating health information systems, and coordinating with nursing staff to track patient flow, and identifying scheduling bottlenecks like overbooking in order to help reduce wait times and improve overall patient experience.
- Gained hands-on experience using 10+ types of clinical equipment including sphygmomanometer, pulse oximeters, glucometers, centrifuges, phlebotomy kits, and vaccine refrigeration units, while also assisting with blood testing procedures and observing proper vaccine storage, preparation, and administration protocols.

New Jersey Institute of Technology

Newark, NJ

Research Intern

May 2024 – August 2024

- Explored countless biomedical engineering concepts and ideas through inspection of drug delivery devices, reviewing mechanical stress and fluid flow tests, and learning how principles of materials science, fluid dynamics, and human physiology influence device performance, dosage accuracy, and overall patient safety outcomes.
- Used advanced 3D modeling software to analyze and refine medical device prototypes, evaluating design and drug features for compliance with FDA regulations, ISO standards, and industry guidelines related to performance, biocompatibility, risk management, and manufacturability requirements.

Crumbl Cookies

Morris Plains, NJ

Crew Member

June 2024 – May 2025

- Processed and managed orders for 150+ customers per shift using POS and ticketing systems, ensuring accurate order entry, efficient workflow coordination, consistency in the product, and timely preparation during busy periods.
- Supported daily operations by assisting customers, resolving order issues, training 50+ employees, monitoring workflow, and performing managerial duties such as coordinating staff and maintaining quality standards.

EXTRACURRICULAR EXPERIENCES

FLL Robotics Mentor

Livingston, NJ

- Mentored a rookie robotics team for 200+ hours in programming, mechanical design, and problem solving, optimizing robot performance, and helping them qualify for the state tournament for the first time in history.

SEWA Volunteering - Presidential Service Award volunteer

Morris County, NJ

- Dedicated 100+ hours to community service initiatives by coordinating with program directors to develop schedules, recruit and train volunteers, manage logistics, and execute large scale outreach and service events.

SKILLS, ACTIVITIES & INTERESTS

Languages: English (fluent), Telugu and French (Conversational Proficiency)

Technical Skills: Python, Data Analytics, Circuitry, CAD, Woodwork, Metalwork, SQL

Activities: Morris Hills Basketball (4 years), JSA (Secretary), Biomedical Engineering Society, Real estate club

Hobbies: College Basketball, Cooking, Weightlifting, Indian Action Movies, and Content Creation